

MILAD SHIRANI

Scientific AI Researcher · Physics AI · Differentiable Scientific Computing

U.S. Permanent Resident (Green Card)

Postdoctoral Associate · Yale University · New Haven, CT

milad.shirani@yale.edu ·  ·  ·  · 

PROFILE

Scientific AI researcher and ML engineer with 10+ years combining continuum mechanics, numerical simulation, differentiable programming, and machine learning for nonlinear solids and coupled multiphysics. Develops physics-constrained neural networks, neural operators, and differentiable scientific-computing systems that embed governing equations, variational principles, conservation laws, and thermodynamic consistency directly into computation. Built data-free GNN solvers that minimize FEM weak-form residuals, matrix-free differentiable finite-element solvers using autodiff and Newton–Krylov methods, and differentiable meshless peridynamics for fracture and delamination. Demonstrated an $11.6\times$ GPU tangent-product speedup for differentiable FEA, $8\text{--}10\times$ speedup for meshless mechanics, and $35\text{--}250\times$ lower neural-operator query cost than benchmark FEM, with 0.05–3% agreement against converged finite-element solutions across coupled systems. Open-source developer with startup experience translating scientific ML methods into engineering systems.

TECHNICAL SKILLS

Physics AI / ML	Physics-constrained neural networks; graph neural networks; neural operators (DeepONets, FNOs); PINNs; surrogate modeling; variational energy-minimizing networks; weak-form / Galerkin residual training
Differentiable Computing	Automatic differentiation; Jacobian-vector products (JVPs); matrix-free nonlinear solvers; differentiable FEA; differentiable meshless mechanics; implicit differentiation; adjoint methods
Numerical Methods	Finite Element Methods (FEM); Natural Element Method (NEM); Peridynamics; Newton–Raphson; Newton–Krylov; dynamic relaxation; weak-form discretization; stability analysis
Scientific Programming	PyTorch, JAX, TensorFlow, Scikit-learn; Python; MATLAB
Physics	Nonlinear elasticity; coupled multiphysics; diffusion–mechanics; thermoelasticity; electrodynamics; generalized continua; Cosserat / micropolar / higher-gradient mechanics; liquid crystal elastomers; hydrogels; polymers
Simulation	FEniCSx; COMSOL Multiphysics
Engineering / MLOps	End-to-end ML pipelines; data ingestion; training and checkpointing; evaluation; modular JAX/PyTorch packages; multi-seed experiments; open-source development; surrogate benchmarking against converged FEM

RESEARCH & PROFESSIONAL EXPERIENCE

Postdoctoral Associate — Yale University

Sept 2023 – Present

Department of Biomedical Engineering · New Haven, CT

- **Matrix-free differentiable FEA — Newton–Krylov:** Built an [open-source](#) PyTorch finite-element solver for large-strain hyperelasticity that eliminates global stiffness-matrix assembly. Residuals and Jacobian–vector products are computed through automatic differentiation; each Newton step is solved with preconditioned Krylov iteration using only matrix-free Jacobian–vector products. Computed exact design gradients through implicit differentiation and an adjoint solve. Achieved $< 10^{-13}$ relative agreement with FEniCSx and an $11.6\times$ GPU speedup for tangent products, solving 10,000 elements in 2.9 s. [Project page](#).
- **Differentiable meshless mechanics — fracture and delamination:** Developed an autodiff-based meshless peridynamics solver for nonlinear solids, including brittle fracture, pressure-driven hydraulic fracture, and Mode-I / mixed-mode delamination without remeshing. Achieved $8\text{--}10\times$ runtime speedup and 9.7×10^{-11} agreement with a validated reference solution. Engineered a matrix-free dynamic-relaxation scheme to overcome Newton–Raphson tangent singularities and converge through unstable crack propagation. [Project page](#).
- **Data-free physics-constrained GNNs for multiphysics:** Developed geometry-informed graph neural networks that solve coupled PDE systems as implicit time-steppers by minimizing the FEM weak-form residual directly, without labeled solution data. Embedded conservation laws and thermodynamic consistency in the computational formulation. The graph

shares the FEM discretization, shape functions, quadrature, and boundary conditions, making the weak-form residual both the training objective and an intrinsic physics-based error monitor. Implemented in JAX and PyTorch with residual parity to $\sim 10^{-12}$.

- **PI-GCN — chemoelasticity:** Developed a graph-convolutional network for transient finite-strain chemoelasticity with four coupled fields: displacement, pressure, chemical potential, and concentration. Used block-weighted weak-form residuals and Adam $\rightarrow$ L-BFGS optimization with FEM and meshless NEM back ends. Achieved 0.24% displacement and 0.73% concentration error over 50 implicit time steps.
- **PI-EPD-GNN — thermoelasticity:** Developed a MeshGraphNets-style Encode–Process–Decode model for transient heat conduction coupled to finite-deformation elasticity. Used Taylor–Hood elements, a Voronoi message-passing graph, and tangent-based loss preconditioning. Achieved 0.14% displacement and 0.011% temperature error.
- **Physics-constrained neural operators and variational networks:** Designed label-free energy-minimizing PINNs and DeepONets in PyTorch and JAX for liquid crystal elastomers, coupling finite-strain deformation and director fields. Trained over FEM/NEM discretizations using total potential energy, Gauss quadrature, and exactly enforced boundary conditions. A single operator predicted full fields in ~ 0.5 – 0.7 ms, with 35 – $250\times$ lower query cost than warm FEM on a 156-node benchmark. Achieved 0.7% shear and 1.6–3% uniaxial displacement error on unseen stretches. Validated quasiconvexity, rank-one convexity, and Legendre–Hadamard conditions. [Project page](#).
- **Inverse constitutive modeling:** Developed PINN-based inverse frameworks in JAX and PyTorch for learning polyconvex stored-energy functions for isotropic and anisotropic materials, including fibrous composites and biological tissues, directly from FEA-generated datasets, with polyconvexity enforced as a hard architectural constraint.
- **PI-FNO — open-source Physics-Informed Fourier Neural Operator:** Designed, implemented, and open-sourced [PI-FNO](#) for 5-class valvular heart-disease classification from phonocardiograms. Embedded cardiac periodicity and S1/S2 frequency hierarchy as hard architectural priors and used a learnable frequency mask initialized from cardiac acoustics. Achieved 97.5% test accuracy across 1,000 recordings. [Project page](#).
- **Multiphysics simulation infrastructure:** Built FEM and multiphysics frameworks using FEniCSx, COMSOL, JAX, and PyTorch for gel swelling, thermoelasticity, and fibrous biomaterials, providing high-fidelity references for surrogate training and benchmarking. Currently extending the hydrogel swelling model to biological tissues and to delamination of layered hydrogel systems.
- **Research leadership:** Led a multidisciplinary team of four researchers across Yale, Brown, and Stanford; mentored PhD and undergraduate researchers in PyTorch-based constitutive-model discovery from biaxial experimental data.
- **Research output:** 3 published · 4 under review · active peer reviewer.

ENTREPRENEURIAL EXPERIENCE

Co-Founder & ML Engineer — UC Berkeley SkyDeck / CITRIS Foundry

Jul 2022 – Sept 2023

Berkeley, CA

- Co-founded an AI startup for cardiovascular disease detection from phonocardiograms; accepted into UC Berkeley SkyDeck Pad-13 and CITRIS Foundry.
- Developed and benchmarked CNN and transfer-learning models for multi-class heart-sound classification, achieving **95% accuracy** on held-out test data and meeting the startup’s clinical deployment target. Owned the end-to-end ML lifecycle, including data preprocessing, model development, validation, benchmarking, and evaluation under a startup development timeline.
- Mentored an undergraduate student in building a hardware digital stethoscope paired with the deployed model.

PH.D. RESEARCH — UNIVERSITY OF CALIFORNIA, BERKELEY

Ph.D. Researcher — University of California, Berkeley

Aug 2016 – Aug 2023

Department of Mechanical Engineering · Minors: Mathematics, Dynamics · Berkeley, CA

- **Graduate textbooks (2):** Co-authored [Principles of Continuum Mechanics](#) (World Scientific) and [Lecture Notes on the Theory of Plates and Shells](#) (Springer).
- Derived quasiconvexity, rank-one convexity, and Legendre–Hadamard inequalities for Cosserat elasticity, fibrous composites, and liquid crystals, providing mathematically rigorous stability and validation criteria for computational and neural predictions.
- Developed constitutive models for fiber-reinforced composites and metamaterials incorporating size-dependent Cosserat and strain-gradient behavior; implemented FEA in COMSOL and Python.

- Developed Kirchhoff-rod models for fibrous materials, fiber-reinforced thin-structure models for heart valves, and computational models of lipid bilayers reproducing biconcave red-blood-cell geometry.
- **Output:** 2 books · 22 peer-reviewed papers · 5 book chapters · Paul Naghdi Fellowship · Bob Steidel Fellowship · **Outstanding Graduate Student Instructor Award**.

EARLIER RESEARCH EXPERIENCE

Graduate Researcher — Pennsylvania State University

Aug 2014 – May 2016

Department of Engineering Science and Mechanics · State College, PA

- Experimental characterization of iron-based shape-memory alloys (DSC, tensile testing) and carbon-fiber/fiberglass composites; photoelastic stress analysis.

Graduate Researcher — Isfahan University of Technology

Sept 2010 – Jan 2013

Department of Mechanical Engineering · Isfahan, Iran

- Proved thermodynamic inconsistency of the classical Brinson (1993) shape-memory-alloy model and derived a corrected constitutive framework validated against experimental data in MATLAB, Python, and Abaqus.
- Derived a thermodynamically consistent magnetic shape-memory-alloy model; a single calibration reproduced 12 distinct magneto-mechanical loading cases. Predicted from thermodynamic reasoning that biaxial compression must produce hysteresis, then designed and executed experiments confirming the prediction.
- **Output:** 8 peer-reviewed papers · **Best Thesis Award (across all degree levels: BSc, MSc, PhD)**.

EDUCATION

Ph.D., Mechanical Engineering — University of California, Berkeley

2023

Minors: Mathematics & Dynamics · Paul Naghdi Fellowship · Bob Steidel Fellowship · Outstanding Graduate Student Instructor Award

M.Sc., Mechanical Engineering — Isfahan University of Technology

2013

Best Thesis Award (across all degree levels: BSc, MSc, PhD) · 8 publications

B.Sc., Mechanical Engineering — Isfahan University of Technology

2009

PUBLICATIONS & OUTPUT

- **Books (2):** *Principles of Continuum Mechanics* · *Lecture Notes on the Theory of Plates and Shells*
- **33** peer-reviewed journal papers (8 MSc · 22 PhD · 3 postdoc) · 4 under review · 5 book chapters
- **Full publication list:** [Google Scholar](#)